Sparse Similarity-Aware Label Smoothing via Learned Latent Spaces

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Abstract

Label smoothing is a widely used regularization technique for reducing overconfidence in neural networks, improving generalization and calibration for multi-class classification. However, conventional uniform label smoothing distributes probability mass indiscriminately across non-target classes, assuming equal severity for all misclassifications and producing increasingly diffuse targets that yield diminishing per-class gradients as the number of classes grows. To address these limitations, we learn a latent space representation and use inter-class distances to identify semantically related classes. Label smoothing redistribution is then restricted sparsely to a small neighborhood of the target class, with soft targets weighted by distance to focus on plausible misclassifications. Experiments demonstrate improved calibration, as measured by the expected calibration error, and suggest that sparsity is not a secondary design choice but a central ingredient for effective label smoothing.

1. Introduction

Misclassifying a truck as a cat is clearly wrong. So is calling it a car. But are these mistakes equally severe? Standard classification losses implicitly treat all mistakes as equally distant: any incorrect class is penalized identically, ignoring semantic or visual similarity. However, one can intuitively argue that real-world label spaces are structured—classes lie on a spectrum of relatedness, and uncertainty is typically concentrated among plausible alternatives rather than spread uniformly across unrelated categories. Ignoring this inherent structure leads to target distributions that fail to reflect these meaningful relationships between classes, which can in turn contribute to poorly calibrated predictive confidence. This observation motivates a simple question: should training objectives encode not only correctness but also the *similarity structure* of the label space?

Modern deep neural networks trained with one-hot labels are capable of achieving high top-1 accuracy. However, they often produce probability estimates that are misaligned with the empirical correctness(Guo et al., 2017). In practical applications, this miscalibration gap provides unreliable confidence estimates, and can contaminate downstream decision-making. This presents an especially significant problem for risk-aware systems that depend on well-calibrated probabilities rather than only on argmax accuracy. For instance, a social media platform employing an overconfident automated content moderation system would ban and restrict a substantial amount of innocent users despite low true certainty of their violations. This gap has motivated extensive work on calibration metrics and procedures, such as reliability diagrams and expected calibration error (ECE), as well as post-hoc methods such as temperature scaling (DeGroot & Fienberg, 1983; Niculescu-Mizil & Caruana, 2005; Naeini et al., 2015; Guo et al., 2017).

Perhaps the most widely adopted training-time remedy is *label smoothing*, which replaces one-hot targets with a convex combination of the hard label and a prior distribution over classes, typically uniform (Szegedy et al., 2015). By discouraging sharply peaked predictive distributions, label smoothing acts as an effective regularizer to mitigate overconfident predictions, and is thus a standard in many high-performing pipelines. Beyond generalization, label smoothing has also been shown to improve confidence calibration in a variety of settings (Müller et al., 2019). Closely related perspectives interpret label smoothing as encouraging higher-entropy outputs or penalizing overconfident predictions (Pereyra et al., 2017). More broadly, several training strategies that mix or soften labels—such as mixup—attribute part of their calibration benefit to the implicit label smoothing effect (Thulasidasan et al., 2019).

However, the canonical form of label smoothing makes a strong modeling assumption: all misclassifications are equally severe, leading to uniform redistribution of the smoothing mass across all incorrect classes (Szegedy et al., 2015). For many recognition problems, however, classes are not equally confusable. A model that is uncertain between

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semantically or visually similar categories (e.g., fine-grained species) is arguably less problematic than one that spreads uncertainty onto unrelated classes. Recent work, such as by Maher & Kull (2021), has explored *class-similarity* smoothing schemes that allocate probability mass in proportion to inter-class similarity rather than uniformly, with the explicit goal of improving calibration. However, is it necessary for smoothing to assign non-zero probability to every incorrect class? This idea motivates an appealing principle: when we smooth labels, we should smooth them *toward plausible alternatives only*.

In this work, we formalize and operationalize this principle via **sparse similarity-aware label smoothing (SSA-LS)**. Rather than redistributing smoothing mass over all non-ground-truth classes, we allocate the smoothing budget only across a small neighborhood of classes most similar to the ground-truth label. This yields a *sparse* target distribution that preserves the regularization benefit of soft labels while avoiding the unrealistic assumption that all incorrect classes are equally plausible. Because the smoothing mass is concentrated on a small set of related classes, the resulting target magnitudes are larger for meaningful alternatives, inducing stronger gradient signals on semantically related alternatives instead of distributing weak gradients uniformly across unrelated classes. Our design is intentionally *drop-in*: it can be used with standard architectures and training recipes without modifying the classifier or loss beyond the target construction.

We evaluate SSA-LS on multiple image classification benchmarks and architectures under controlled training conditions, measuring both confidence calibration and predictive performance. Across dataset–architecture pairs and smoothing strengths, SSA-LS consistently reduces calibration error relative to standard uniform label smoothing while preserving comparable accuracy and negative log-likelihood. We further conduct targeted ablations that isolate the role of sparsity, clarifying how neighborhood structure influences the calibration–accuracy trade-off.

2. Related Work

The miscalibration of modern neural networks has been extensively documented (Niculescu-Mizil & Caruana, 2005; Guo et al., 2017). Guo et al. show that contemporary deep models are often systematically overconfident and demonstrate that temperature scaling — a simple post-hoc rescaling of logits — can substantially improve calibration without affecting accuracy. While effective, post-hoc methods operate only on the final predictions and do not influence the training objective or learned representations.

Training-time approaches such as label smoothing address calibration directly by modifying supervision. Subsequent work has extended this idea beyond uniform redistribution. Online label smoothing (Zhang et al., 2021) dynamically constructs soft targets during training using model statistics, allowing the smoothing distribution to adapt over time rather than remaining fixed. More recent methods incorporate class structure when designing smoothing distributions, allocate probability mass according to relationships between classes. In particular, Liu & JáJá (2020) leverage autoencoder latent spaces and word embeddings to define class-based soft targets. Variational label smoothing (Yang et al., 2025) further connects latent variable modeling with adaptive supervision, showing that uncertainty in latent representations can naturally induce soft labeling behavior. These works highlight the importance of representation geometry when redistributing label mass and move beyond purely uniform or prediction-based smoothing strategies.

Despite these advances, existing approaches primarily focus on how smoothing mass should be distributed, not on *where* it should be distributed. To the best of our knowledge, prior work does not explicitly investigate sparsity as a design principle in label smoothing. Our work introduces sparsity as a first-class constraint: rather than spreading probability mass across all classes or dense semantic neighborhoods, we restrict redistribution to a small set of nearest classes in a learned latent space. This perspective reframes label smoothing as a structured sparse allocation problem and constitutes the central conceptual contribution of our method.

3. Preliminaries

Let $\mathcal{X}$ and $\mathcal{Y} = \{1, 2, \ldots, C\}$ denote the input and output spaces respectively, where C is the number of classes present. We train neural network $h_\theta : \mathcal{X} \to \mathbb{R}^C$ that outputs logits $z = h_\theta(x) = (z_1, z_2 \ldots z_c)$ for input $x \in \mathcal{X}$. Here, the model's predictive distribution is then

$$p_\theta(y = c|x) = \mathrm{softmax}_c(h_\theta(x)) = \frac{\exp(z_c)}{\sum_{i=1}^{C} \exp(z_i)}$$

Let $q(y|x)$ denote the target distribution. The cross-entropy between the two distributions q and p_θ is then defined as

$$H(q, p_\theta) = -\sum_{c=1}^{C} q(c|x) \log p_\theta(c|x)$$

The standard one-hot classification target distribution is defined as $q(c|x) = \mathbb{1}(c = y)$. So, the cross-entropy loss for a single sample becomes $\mathcal{L}_{\mathrm{CE}}(x, y; \theta) = -\log p_\theta(y|x)$. Over n samples, we then have

$$\mathcal{L}_{\mathrm{CE}}(\theta) = -\sum_{i=1}^{n} \log p_\theta(y_i|x_i) = \mathrm{NLL}(\theta)$$

where NLL refers to the negative log likelihood. Here, observe that

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_c} = \begin{cases} p_\theta(c|x) - 1 \text{ if } c = y \\ p_\theta(c|x) \quad \text{otherwise} \end{cases}$$

Essentially, for one-hot labels, gradients push $p_\theta(c|x)$ to 0 for incorrect classes and 1 for correct class y. This push towards extreme probabilities leads to an overconfident neural network, inducing the need for confidence calibration.

A classifier is perfectly calibrated if

$$\mathbb{P}(\hat{Y} = y|\hat{P} = p) = p, \;\; \forall p \in [0, 1]$$

where $\hat{Y} = \arg\max p_\theta(c|x)$ is the predicted class and $\hat{P} = \max p_\theta(c|x)$ the associated confidence. Essentially, it means that among all predictions made with confidence p, the model is correct $100p\%$ times.

3.1. Label Smoothing

Label smoothing is the practice of replacing the one-hot target distribution with a softened target. Here, let the smoothing factor ϵ denote the probability mass that is redistributed among incorrect classes. Then, we have target distribution

$$q_\epsilon(c|x) = (1 - \epsilon)\mathbb{1}(c = y) + \epsilon \cdot s_y(c|x)$$

where $s_y(c|x)$ is a valid probability distribution s.t. $s_y(y|x) = 0$.

Uniform label smoothing is the standard and simple method where ϵ is redistributed uniformly over all incorrect classes. Specifically, we have

$$q_\epsilon(c|x) = \begin{cases} 1 - \epsilon \text{ if } c = y \\ \dfrac{\epsilon}{C - 1} \text{ otherwise} \end{cases}$$

3.2. Metrics

Expected calibration error, a useful scalar summary statistic of calibration (Guo et al., 2017), is the expectation of difference between the model's confidence and accuracy

$$\mathbb{E}_{\hat{P}}\left[\left|\mathbb{P}\left(\hat{Y} = y|\hat{P} = p\right) - p\right|\right]$$

As formulated by Naeini et al. (2015), and borrowing notation from Guo et al. (2017), to approximately compute ECE, we partition predictions into M equal intervals. Let interval $I_m = \left(\frac{m-1}{M}, \frac{m}{M}\right]$ and B_m be the corresponding set of indices of samples based on prediction confidences. Then, we define

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{1}(\hat{y}_i = y_i)$$

and

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i$$

where $\hat{y}_i$ is the predicted label with associated confidence $\hat{p}_i$ for sample x_i with true label y_i. Finally, we may approximate ECE with the interval bins as the weighted average

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$$

where n is the total number of samples. Observe that for a perfectly calibrated model we have $\text{acc}(B_m) = \text{conf}(B_m)$ for all m i.e. $\text{ECE} = 0$. This is the primary metric we employ to compare performance of label smoothing methods ahead.

We also consider maximum calibration error (Naeini et al., 2015) defined as

$$\text{MCE} = \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$$

as an additional metric. For visual aid, we rely on reliability diagrams (DeGroot & Fienberg, 1983; Niculescu-Mizil & Caruana, 2005).

3.3. β-VAE

The variational autoencoder (Kingma & Welling, 2022) is a deep generative model that models data $x \in \mathcal{X}$ using latent variables $z \in \mathcal{Z}$. It consists of an encoder $q_\phi(z|x)$ and a decoder $p_\theta(x|z)$ connected through a probabilistic latent space, typically of much smaller dimension than the data and modeled after a chosen prior distribution. The encoder maps each sample point to a distribution, in the latent space; conversely, the decoder maps points of the latent space to the input space. They are jointly trained to maximize the evidence lower bound (ELBO)

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x|z)\right] - \text{KL}\left(q_\phi(z|x) \| p(z)\right)$$

where $p(z)$ is a prior over the latent variables. Observe that the first term encourages improved reconstruction quality, while the other term encourages matching the latent space to the prior distribution.

The encoder $q_\phi(z|x)$ outputs the parameters of a conditional latent distribution, usually a Gaussian parameterized by mean $\mu_\phi(x)$ and variance $\sigma^2_\phi(x)$. The learned latent space is fully specified by the encoder parameters and the assumed prior distribution, and thus can be accessed independently of the decoder.

The $\beta-$VAE (Higgins et al., 2017) introduces factor $\beta > 0$ to scale the KL-divergence term.

$$\mathcal{L}_{\beta-\text{VAE}} = \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x|z)\right] - \beta\text{KL}\left(q_\phi(z|x) \| p(z)\right)$$

Increasing β prioritizes the prior matching objective i.e. the constraint that the encoder $q_\phi(z|x)$ match the prior $p(z)$. This encourages a more structured and isotropic latent space, and can lead to more disentangled latent representations.

4. Method

4.1. Overview

We propose a class-wise similarity-aware training target distribution based on representations in a learned latent space. First, a β-VAE is trained to obtain a structured latent representation of the data, and its encoder is subsequently fixed as the feature extractor. Class-wise latent statistics are then computed by aggregating latent representations of all samples belonging to the same class. In particular, distances between classes in the latent space are established. For each class, these distances are used to identify a fixed number of plausible misclassifications. Table 1 shows representative examples of nearest neighbors on CIFAR-100 discovered by our method.

Table 1. Representative examples of top-5 nearest neighbors in the learned latent space on CIFAR-100

CLASS	TOP-5 NEAREST NEIGHBORS
boy	girl, man, woman, baby, table
dolphin	shark, whale, turtle, ray, seal
oak tree	pine tree, maple tree, willow tree, palm tree, tractor
rose	tulip, poppy, orchid, lobster, crab
tiger	leopard, fox, lion, snake, rabbit

Then, the smoothing weight is redistributed among these specific incorrect classes in a weighted manner. Figure 1 visualizes the resulting sparse target distribution for an example class, highlighting how the probability mass is concentrated on a small set of similar alternatives rather than spread densely or uniformly. This procedure defines a sparse and smoothed target distribution over classes that is used to train the neural network.

While we employ a β-VAE trained from scratch on the target dataset to obtain an unsupervised embedding that reflects dataset-specific structure, our goal is not to optimize the similarity estimator itself but to isolate the effect of sparsity in label smoothing. The method formulation itself is agnostic to the choice and can be combined with alternative similarity estimators instead. Ahead, we refer to our method as Sparse Similarity-Aware Label Smoothing (SSA-LS).

4.2. Class-wise Latent Distances

The first step to obtain class-wise similarity-aware targets is to establish which classes are similar to each other and quantify how similar they are. Variational autoencoders

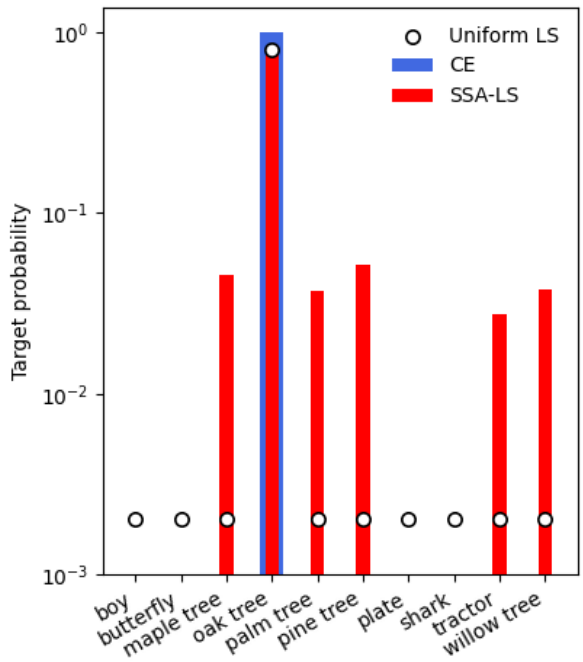


Figure 1. Comparison of example training targets for "oak tree" CIFAR-100 class, under CE, uniform label smoothing, and our method SSA-LS. Plot, inspired by Yang et al. (2025), depicts the target class and its 5 closest neighbors along with 4 random classes.

(VAEs) learn a continuous, probabilistic latent space with semantic continuity (Kingma & Welling, 2019) i.e. similar inputs tend to be embedded nearby, enabling meaningful geometric reasoning based on latent representations. Prior work has demonstrated that this latent structure supports clustering, with latent proximity aligning with semantic similarity even without supervision, as evidenced by methods such as Variational Deep Embedding (Jiang et al., 2016). Thus, we may use distance-based analysis on the latent space to determine semantically similar classes. Improving on this, β-VAEs strengthen regularization towards modeling the latent space after the prior, encouraging more smoothly structured and disentangled latent representations (Higgins et al., 2017), leading to a more semantically meaningful latent space. The disentangled latent structure allows for more stable and interpretable latent space distance-based comparisons for downstream analysis.

So, we first train a β-VAE on the training data and subsequently freeze all its parameters. Using the VAE's encoder, we collect latent representations of the input dataset. To ensure deterministic nature, we only consider the mean $\mu_\phi(x)$ outputted by the encoder of the distribution that sample x is mapped to. Let $d(i, j)$ represent the distance between

classes C_i and C_j. We define

$$d(i,j) = \|\mu_i - \mu_j\|_2, \text{ where } \mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} \mu_\phi(x) \quad (1)$$

i.e. the centroid distance between the two classes.

4.3. Similarity-Aware Target Construction

Here, we introduce the sparsity factor k. In simple words, to define softened class-wise targets for our neural network, we redistribute smoothing factor ϵ among the k closest classes to the target class (excluding itself). Let $\mathcal{N}_k(c)$ denote the set containing the k closest classes to class c, as determined by the distance metric. We define vector $r(y)$ s.t.

$$r(y)_c = \begin{cases} d(y,c) \text{ if } c \in \mathcal{N}_k(y) \\ \infty \text{ otherwise} \end{cases}$$

Note that $r(y)$ depends on k, but we leave it out for simpler notation. Then, for given ground-truth label y, we set the target distribution

$$q_\epsilon(c|y) = \begin{cases} 1-\epsilon \text{ if } c = y \\ \epsilon \cdot \text{softmin}_c(r(y);T) \text{ if } c \in \mathcal{N}_k(y) \\ 0 \text{ otherwise} \end{cases}$$

where T is the temperature scaling parameter controlling the sharpness of the softmin function (Guo et al., 2017). Larger values of T render a more uniform distribution of the smoothing mass ϵ among classes in $\mathcal{N}_k(y)$; as $T \to 0$, the mass collapses on the nearest class. Since the sparsity factor k already restricts the redistribution to the desired number of classes, very low values of T become redundant. In practice, we therefore select T to achieve a moderately non-uniform redistribution of ϵ among classes in $\mathcal{N}_k(y)$.

Observe that, for simplicity, we opt for a target distribution independent of the input sample x. An interesting direction worth exploring is to account for the relationship between x's latent representation and its class's centroid as a measure of how similar x is to its nearest neighboring classes, which we leave for future work.

Given the similarity-aware target distribution defined above, the model is trained using standard cross entropy loss. Other aspects of training, including the architecture, learner rate schedule, etc. follow conventional supervised training.

5. Experiments & Analysis

5.1. Setup

5.1.1. Datasets

The CIFAR-100 (Krizhevsky & Hinton, 2009) dataset contains 32×32 color images from 100 classes. It contains a total of 50,000 train and 10,000 test images. The Tiny ImageNet dataset (Deng et al., 2009) contains 64×64 color images from 200 classes, with 100,000 training images and 10,000 validation images. For data augmentation, we apply random cropping and horizontal flipping to the CIFAR-100 and TinyImageNet training sets (Krizhevsky et al., 2012). Apart from dataset normalization, no other preprocessing is performed.

5.1.2. Architectures

We train a convolutional β-VAE as per the standard implementation to obtain the latent representations of the images. Following Bowman et al. (2015), the KL-divergence weight β is annealed linearly from 0 to 4 over the first 20% epochs and kept constant thereafter. The latent dimensionality is set to 128 for CIFAR-100 and 64 for TinyImageNet, and the Adam optimizer (Kingma & Ba, 2015) is employed.

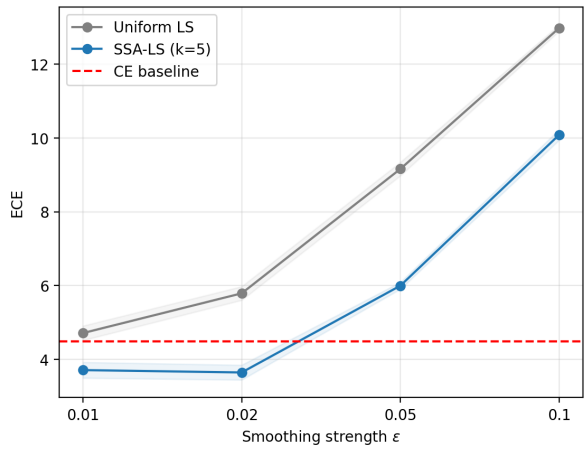
For image classification, we train ResNet-18 (He et al., 2015) and DenseNet-121 (Huang et al., 2016) on CIFAR-100, and ResNet-18 on TinyImageNet. We use the standard architectures as introduced in their respective papers. All models are trained from scratch using the SGD optimizer with Nesterov momentum and a cosine annealing learning rate schedule.

5.2. Calibration Across Smoothing Strengths

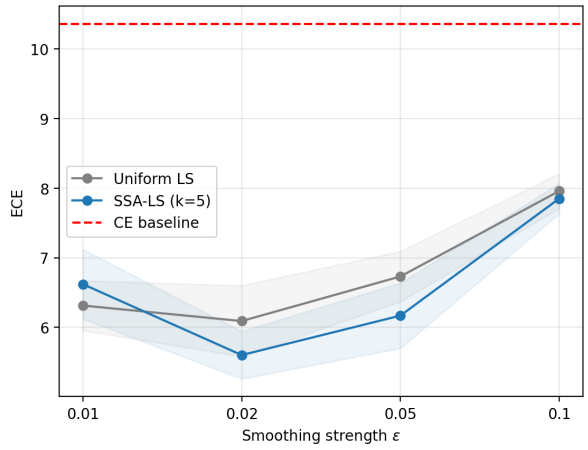
Before evaluating sparsity, we explore whether SSA-LS consistently improves confidence calibration across different smoothing strengths ϵ. We consider models trained with standard one-hot labels (CE) as the baseline, and then evaluate uniform label smoothing (LS) and SSA-LS over $\epsilon \in \{0.01, 0.02, 0.05, 0.1\}$. To isolate the effect of similarity-aware sparsity, we fix sparsity factor $k = 5$ for all SSA-LS models in this section. All models are trained under identical training and optimization settings, ensuring that any differences in performance can be attributed to the choice of training targets. As discussed in Section 3.2, we assess calibration using expected calibration error (ECE) as the primary metric. The maximum calibration error (MCE), negative log-likelihood (NLL) and classification accuracy are reported as complementary measures.

Figure 2 illustrates the behavior of confidence calibration using uniform label smoothing vs. SSA-LS as a function of the smoothing strength ϵ. Across all settings, SSA-LS achieves a lower ECE than uniform label smoothing at the same smoothing strength for a majority of ϵ values considered. This indicates that incorporating similarity-aware sparsity into the softened target distribution consistently improves calibration under identical hyperparameter choice.

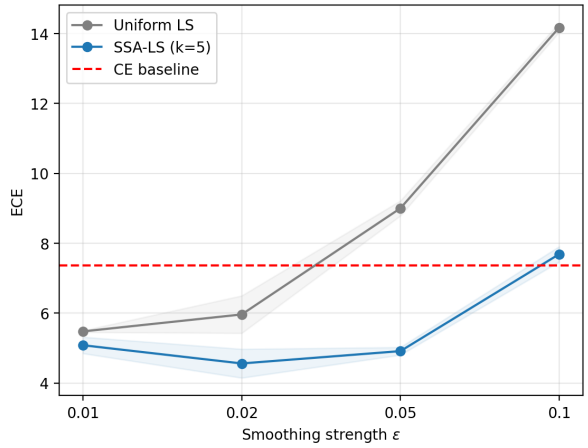
This behavior can be explained by the differences in the construction of the smoothed targets. Redistributing proba-



(a) CIFAR-100, ResNet-18.



(b) CIFAR-100, DenseNet-121.



(c) TinyImageNet, ResNet-18.

Figure 2. Expected Calibration Error (ECE) vs. smoothing strength ϵ for Uniform LS and SSA-LS along with CE baseline across dataset-architecture pairs. Curves show the mean over three seeds; shaded regions denote ± 1 std.

bility mass over all incorrect classes, including semantically unrelated ones, may introduce ambiguity that is less aligned with the underlying class structure. In contrast, SSA-LS instead concentrates the redistributed mass on a small set of similar classes, leading to uncertainty that is more closely aligned with plausible misclassifications.

Both methods exhibit sensitivity to ϵ, though the resulting calibration decay differs in magnitude. In particular, generally increasing ϵ leads to rapid degradation in confidence calibration, and can even lead to a model worse calibrated than one trained with vanilla one-hot targets. This effect is most pronounced for uniform label smoothing at higher values of ϵ, where the ECE sharply rises for ResNet-18 on both CIFAR-100 and TinyImageNet. We observe that SSA-LS helps mitigate this degradation by yielding lower ECE at these large ϵ, providing particularly strong improvements for the ResNet-18 models. However, SSA-LS does not eliminate sensitivity to the smoothing strength entirely. For the DenseNet-121 architecture on CIFAR-100, although SSA-LS achieves lower ECE at the optimal ϵ, its variation across ϵ is comparable to uniform label smoothing, with overlapping results within one standard deviation. It is, however, worth noting that variance in ECE is notably higher for DenseNet-121 than for ResNet-18.

While our work primarily focuses on improving confidence calibration, predictive accuracy remains a crucial consideration. Prior work has shown that regularization methods induce an accuracy-ECE tradeoff (Zhang et al., 2022), where improvements in confidence calibration may come at the expense of classification accuracy. It is thus critical to verify the better calibration achieved by SSA-LS models does not result in a systemic degradation of performance power. We observed that SSA-LS models trained with larger smoothing strengths demonstrated reduced classification accuracy, a behavior consistent with the calibration–accuracy trade-off introduced by stronger regularization. However, these models also exhibit poor calibration and are therefore not selected as optimal under our evaluation. The best-calibrated SSA-LS models maintain competitive, or slightly lower, accuracy and likelihood, indicating that the observed calibration improvements do not arise from significantly sacrificing predictive performance. The maximum calibration error (MCE), test accuracy, and NLL for the optimal model of each training regime, as determined by ECE, are reported in Table 2.

Overall, we observe that the same qualitative trend holds across all dataset-architecture pairs, indicating that the benefit of SSA-LS mass redistribution is not limited to selected settings. These results suggest that similarity-aware sparse label smoothing consistently improves calibration relative to dense label smoothing.

Table 2. Calibration and accuracy comparison using optimal hyperparameters. For Uniform LS and SSA-LS (k=5), the best ϵ is selected based on validation ECE.

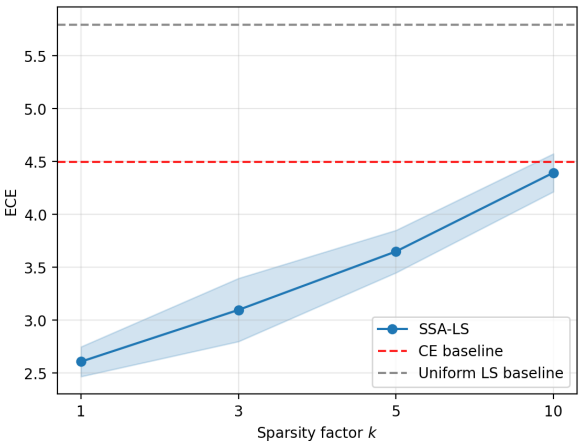
DATASET	MODEL	METHOD	ϵ	ECE↓	ACC↑	NLL↓	MCE↓
CIFAR-100	RESNET-18	CE	–	4.50	78.32	0.87	**12.48**
		UNIFORM LS	0.01	4.72	**79.07**	0.90	18.19
		SSA-LS	0.02	**3.65**	78.86	**0.86**	14.34
CIFAR-100	DENSENET-121	CE	–	10.36	**67.88**	1.33	20.99
		UNIFORM LS	0.02	6.09	67.70	1.35	15.79
		SSA-LS	0.02	**5.60**	67.34	**1.30**	**13.38**
TINYIMAGENET	RESNET-18	CE	–	7.36	**66.16**	**1.53**	17.61
		UNIFORM LS	0.01	5.48	65.56	1.56	12.04
		SSA-LS	0.02	**4.56**	65.18	1.57	**10.67**

5.3. Effects of Sparsity

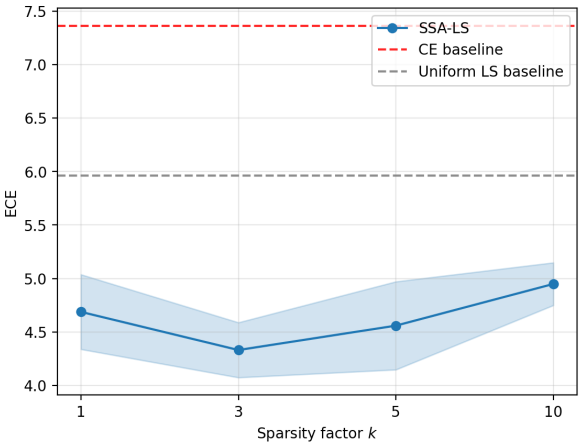
While our sparse label smoothing protocol redistributes the probability mass among similar classes, an implicit but critical design choice is the number of such classes that receive this mass i.e. the sparsity factor k. In this section, we argue that sparsity is not a secondary hyperparameter but instead a defining characteristic of effective label smoothing. We demonstrate that restricting the redistribution to a small neighborhood is crucial for achieving improved confidence calibration.

Redistributing the smoothing mass across many classes weakens the locality assumption underlying similarity-aware smoothing. This effect is particularly pronounced for datasets with a large number of classes, where dense similarity-aware smoothing risks collapsing into uniform label smoothing, as variations in mass allocation become negligible relative to the true class mass $1 - \epsilon$. Preserving meaningful differences in such settings requires extreme tuning of mass reallocation, often enforcing probability gaps across multiple orders of magnitude; in practice, such tuning yields target distributions that are effectively sparse, in spite of being nominally dense.

Even within SSA-LS, increasing the sparsity factor k causes the target distribution to progressively resemble uniform label smoothing, thus ignoring semantic distinctions. Conversely, overly restrictive sparsity may underutilize neighborhood information and amplify sensitivity to errors in the latent representation employed to determine similar classes. Together, these considerations suggest that effective similarity-aware label smoothing requires balancing the preservation of semantic locality and leveraging sufficient available neighborhood information. We therefore hypothesize that moderate sparsity—dependent on the number of classes in the dataset and their inherent underlying structure—yields optimal calibration by maintaining tight meaningful neighborhoods while maximizing the exploitable information within them.



(a) CIFAR-100, ResNet-18 ($\epsilon = 0.02$).



(b) TinyImageNet, ResNet-18 ($\epsilon = 0.02$).

Figure 3. Ablation over sparsity factor k for SSA-LS. Expected Calibration Error (ECE) is shown as a function of k, averaged over three random seeds. Shaded regions denote ± 1 standard deviation. The dashed red line indicates the CE baseline, and the dashed gray line indicates Uniform LS evaluated at the same ϵ.

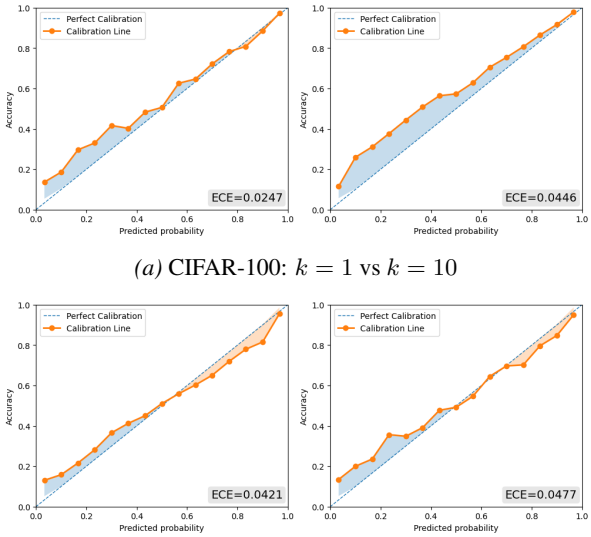
To evaluate this hypothesis, we systematically vary the sparsity factor k used in SSA-LS, while holding all other training conditions fixed. We fix smoothing strength $\epsilon = 0.02$, as was found optimal in the preceding section, and train ResNet-18 models on CIFAR-100 and TinyImageNet. Figure 3 illustrates the effects of varying sparsity on expected calibration error across both datasets.

For CIFAR-100, we observe that ECE monotonically increases with k, with the lowest ECE achieved at $k = 1$. In contrast, TinyImageNet exhibits a non-monotonic relationship between sparsity and ECE, where ECE first decreases to a minimum at $k = 3$ before increasing again as sparsity is further relaxed.

These results highlight the importance of sparsity in label smoothing to further improve confidence calibration. Particularly, we observed that selecting the optimal k reduced ECE by up to 30% for CIFAR-100 and 4% for TinyImageNet as compared to $k = 5$, which was already generally performing better than the dense uniform label smoothing. Moreover, the results illustrate the dataset-dependent nature of optimally effective sparsity. For CIFAR-100, which contains only 100 classes, redistributing the smoothing mass to strictly the closest neighbor appears sufficient to capture all meaningful semantic relationships, with additional neighbors introducing noise and diluting locality. On the other hand, TinyImageNet's larger label space seems to benefit from larger neighborhoods, where multiple related classes provide useful information for smoothing. This supports the view that sparsity should be carefully chosen in accordance with the dataset scale and structure rather than treated as a secondary training hyperparameter.

Figure 4 presents reliability diagrams comparing different sparsity levels k for SSA-LS. In both CIFAR-100 and TinyImageNet, smaller values of k produce calibration curves that more closely follow the perfect-calibration diagonal. In contrast, larger k leads to systematic deviations through all confidence bins, indicating increased miscalibration. This behavior is reflected quantitatively in the lower ECE values observed for smaller k.

The accuracy–calibration trade-off commonly associated with regularization methods remains a potential concern here. However, we observed that classification accuracy remains largely invariant across sparsity levels. On CIFAR-100, top-1 accuracy varied narrowly from 78.72% to 78.97%, while on TinyImageNet it ranged from 65.18% to 65.86%. These fluctuations are comparable to run-to-run variability across random seeds, with pooled accuracy over all k values remaining stable at $78.85 \pm 0.28\%$ and $65.52 \pm 0.30\%$ for CIFAR-100 and TinyImageNet, respectively. This indicates that changes in k primarily affect the confidence calibration rather than the predictive performance. Thus, controlling sparsity is a practical mechanism for reducing ECE with minimal impact on accuracy.



(a) CIFAR-100: $k = 1$ vs $k = 10$

(b) TinyImageNet: $k = 3$ vs $k = 10$

Figure 4. Reliability diagrams comparing sparsity levels k in SSA-LS. Lower k produces improved calibration, notably decreasing calibration deviations throughout the domain. The dashed diagonal indicates perfect calibration.

Together, these results establish that sparsity is a dataset-dependent design choice that governs calibration performance behavior without sacrificing predictive power.

6. Conclusion

We introduced Sparse Similarity-Aware Label Smoothing (SSA-LS), a calibration-focused variant of label smoothing that redistributes probability mass only among semantically similar classes. By leveraging a learned latent space, SSA-LS preserves sparsity while aligning uncertainty with meaningful class relationships. Across CIFAR-100 and TinyImageNet, SSA-LS consistently improves ECE over uniform label smoothing while maintaining comparable accuracy and NLL. Our results show that targeted redistribution is more robust to large smoothing strengths, where uniform smoothing often degrades performance.

Importantly, our ablation studies further show that sparsity plays a critical role in effective calibration: small neighborhoods are sufficient to capture useful class structure, and increasing redistribution beyond this range may yield harm. These findings suggest that calibration benefits arise not from indiscriminate probability spreading, but from structured uncertainty that reflects class similarity.

More broadly, SSA-LS highlights a simple but powerful principle: effective label smoothing must respect the geometry of the label space. Future work may explore adaptive neighborhood selection and class-wise smoothing strengths.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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